RESEARCH PORTFOLIO

Joshua Daniel Upshaw, PhD

Cognitive neuroscience | EEG and ERPs | eye tracking | fMRI | behavioral science | clinical data science and AI

I study how attention, cognitive control, emotion, and decision processes change when people encounter salient information, then build measurement systems that make those changes observable. My work has progressed from behavioral and psychometric research to EEG/ERP, spectral analysis, high-rate eye tracking, secondary fMRI analysis, and large-scale real-world clinical data science.

Research stance
A useful signal is not enough. I want to know what generated it, whether the measurement is valid, which preprocessing and modeling choices change the conclusion, whether effects persist across participants or time, and what next experiment would distinguish competing explanations.

METHODS AND ANALYTICAL RANGE

EEG and ERPs BioSemi ActiveTwo, Net Station, task-locked N2/P2/P3, theta/alpha/beta power, ICA/ICLabel, artifact control.	**Eye tracking** EyeLink 1000 Plus at 1000 Hz, calibration, fixation/AOI, saccades, blinks, pupil dynamics, mixed-effects analysis.	**fMRI and neuroimaging** ABCD secondary analysis, MID task, ROI-based BOLD measures, NAcc/AIns, regression and moderation in RMarkdown.
Behavior and psychometrics Attention, creativity, affect, mindfulness, smartphone use, stress/anxiety, repeated-measures and correlational designs.	**Clinical RWE and ML** Longitudinal EHR data, cohort reconstruction, NLP extraction, rare-outcome prediction, temporal validation, calibration.	**Research governance** IRB/human subjects, ICH-GCP, reproducible pipelines, provenance, data quality, claims calibration, prospective measurement design.

PORTFOLIO EVIDENCE MAP

Primary experimental neuroscience First-author EEG/ERP and spectral EEG studies; doctoral EyeLink dissertation.	**Secondary neuroimaging** ABCD fMRI course project testing screen time, reward processing, and caregiver warmth.
Behavioral and affective science First-author creativity/social media study plus collaborative research on imagination, anxiety, attention, and affect.	**Translational clinical AI** NSF SBIR Phase I retrospective oncology study spanning NLP, ML, workflow timing, and prospective pilot design.

STUDY 1 | EEG, ERP, AND DIGITAL DISTRACTION

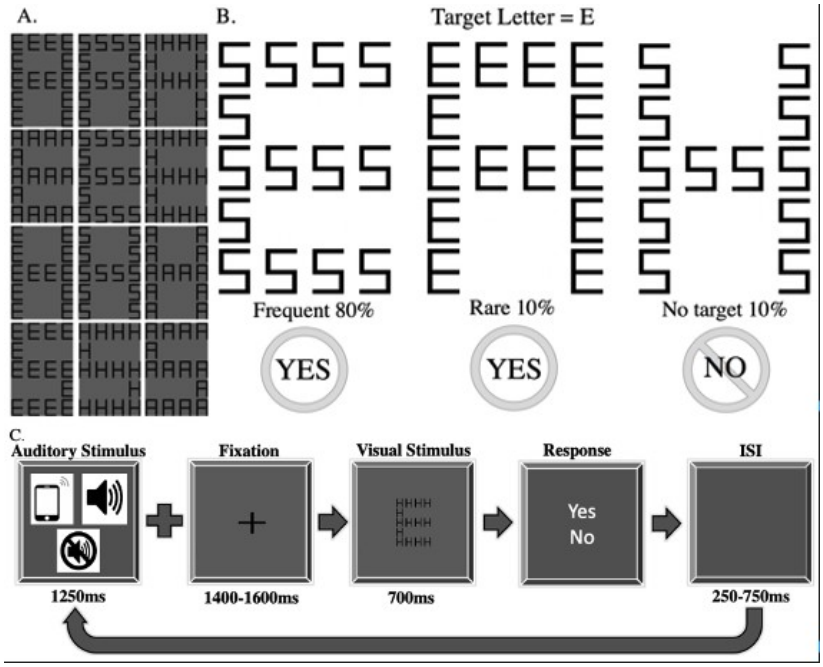
The hidden cost of a smartphone

First-author peer-reviewed study, PLOS ONE, 2022 | Behavioral and electrophysiological cognitive-control experiment

QUESTION

Do smartphone notification sounds alter attention and cognitive control during a visual oddball task, and do individual differences in smartphone-addiction proneness change that response? The study used temporally precise ERPs to separate early attentional processing from conflict monitoring and later stimulus categorization.

73	69	54
participants recruited	behavioral sample	final ERP sample



Local-global Navon oddball paradigm used across the notification experiments. Frequent, rare, and no-target trials were paired with auditory conditions.

ACQUISITION AND ANALYSIS

Experimental structure 960 total trials with frequent, rare, and no-target local/global stimuli; auditory notification and control conditions.	**Neural endpoints** P2, N2, and P3 ERP amplitudes; oddball contrasts used to index attention and cognitive control.
Quality control ICA/ICLabel-assisted artifact correction; participants with more than 25% uncorrectable artifact contamination were excluded.	**Statistical approach** Trial-level regression and condition contrasts across reaction time, accuracy, ERP amplitudes, and smartphone-use individual differences.

WHAT THE DATA SUPPORTED

- Across 37,769 trial observations, rare targets produced a reliable reaction-time oddball effect. Smartphone-sound trials were slightly slower than control-sound trials, 432.97 vs 429.70 ms, a small effect.
- N2 amplitude differed between smartphone and control sound conditions, $t(53) = -2.09$, $p = .041$, while P3 did not differ. The paper interpreted this as greater recruitment of control-related neural activity during notification trials.
- Smartphone-addiction proneness did not consistently moderate behavioral or neural sound effects, an important null result that constrained the interpretation.

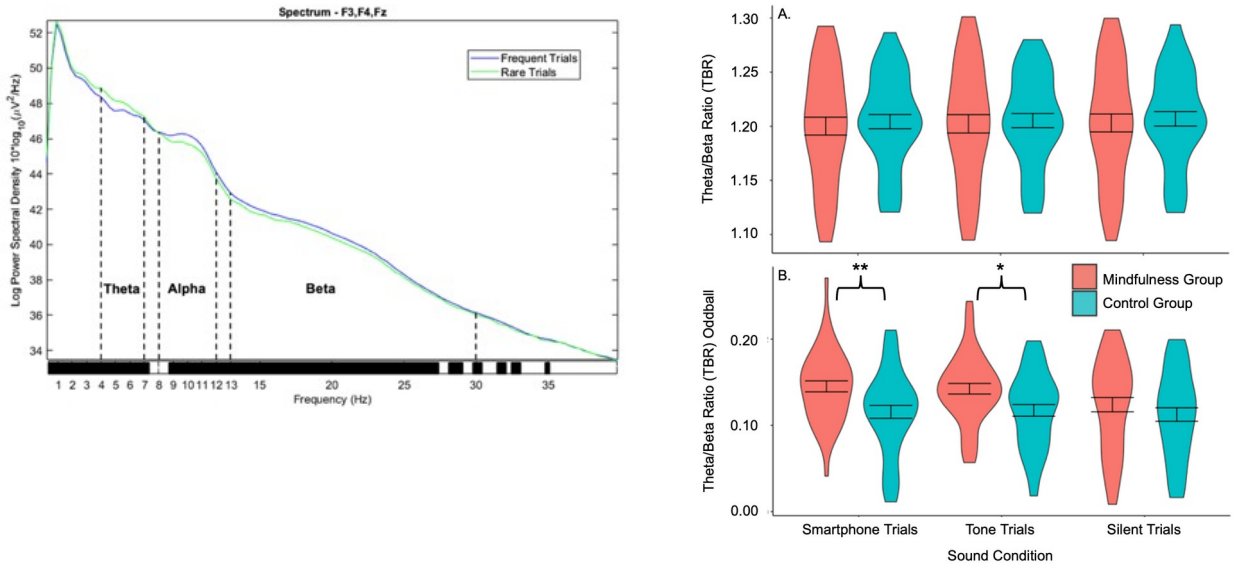
STUDY 2 | SPECTRAL EEG, MINDFULNESS, AND COGNITIVE CONTROL

From auditory perturbation to task-locked spectral dynamics

First and corresponding author, Biological Psychology, 2023 | EEG spectral power and mindfulness induction

This study extended the notification paradigm from ERP components to frequency-domain measures. Participants were randomized to a brief mindfulness induction or neutral narration control, then completed the local-global task while notification, tone, and silence conditions were interleaved within person.

101	87	44 / 43
eligible participants	final EEG sample	mindfulness / control



Published spectral-density and theta/beta-ratio visualizations from the study.

SIGNAL PROCESSING AND QUALITY CONTROL

- 32-channel BioSemi ActiveTwo EEG sampled at 512 Hz, with vertical and horizontal EOG and event synchronization to target onset.
- Downsampled to 256 Hz; high-pass and line-noise processing; bad-channel interpolation; common-average rereferencing; artifact subspace reconstruction; extended Infomax ICA and ICLabel; 35 Hz low-pass filtering.
- The final corrected manuscript reports an average of 2.73 non-brain ICA components flagged per participant, with no group difference in rejected components, t(85) = 0.06, p = .951.
- Frontal spectral analyses quantified theta 4-7 Hz, alpha 8-12 Hz, beta 13-30 Hz, and theta/beta ratio, complemented by ERP analyses.

INTERPRETATION WITH EVIDENTIARY RESTRAINT

Notifications were associated with lower theta and higher alpha/beta power on target trials. The mindfulness group showed stronger task-related differentiation in selected theta/TBR contrasts, particularly on smartphone trials. However, the omnibus group-by-sound TBR interaction was not significant, so the result is best treated as context-sensitive evidence rather than a general claim that a brief mindfulness induction improves cognitive control.

Why this project matters

The project demonstrates a full neural-analysis chain: controlled perturbation, synchronized acquisition, artifact/source separation, spectral decomposition, repeated-measures inference, reviewer-driven error correction, and calibrated interpretation of positive and null findings.

STUDY 3 | EYELINK, EMOTIONAL ATTENTION, AND CROSS-MODAL DISTRACTION

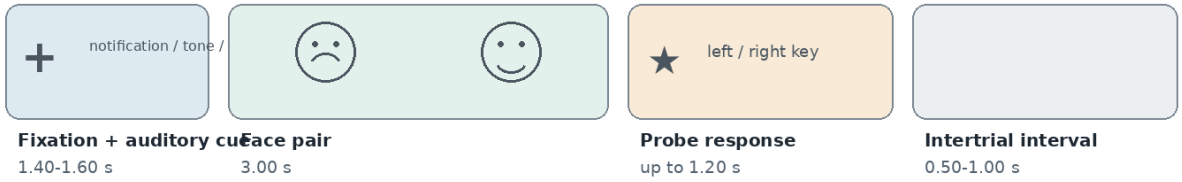
A gaze experiment built around timing, calibration, and multiple physiological readouts

PhD dissertation, University of Arkansas, 2025 | OSF preregistered eye-tracking study

The dissertation asked whether smartphone notification sounds alter visual attention to emotional faces and whether early orienting can be separated from sustained gaze, blinking, pupil dynamics, saccades, and behavioral response. The dot-probe design crossed auditory conditions with angry, happy, and neutral face information.

50	1000 Hz	360
participants enrolled	EyeLink sampling rate	experimental trials

Eye-tracking trial architecture



360 trials | 6 blocks x 60 | 30 emotional face pairs | right-eye EyeLink 1000 Plus at 1000 Hz

Sanitized reconstruction of the trial architecture. Generic icons replace licensed face stimuli.

MEASUREMENT SYSTEM

Calibration and QC 9-point calibration; mean validation error target <=0.75 degrees and maximum <=1.50 degrees; recalibration across blocks.	**Gaze endpoints** Fixation latency/duration/count, first saccade, blink measures, pupil magnitude/variability, reaction time, and accuracy.
Pupil processing Blink-adjacent exclusion and interpolation, baseline correction, standardization, and outlier control. Pupil values retained in EyeLink arbitrary units.	**Modeling** Repeated-measures ANOVA plus generalized and linear mixed-effects models with endpoint-specific quality rules.

Figure 5
Main Effect of Emotion on First Fixation Time

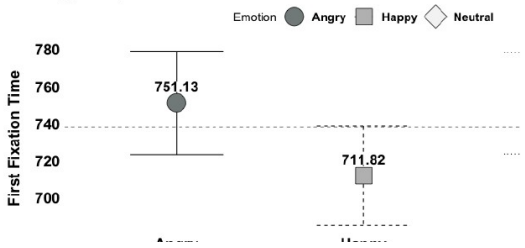
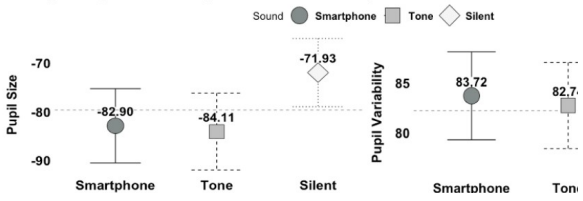


Figure 9
Main Effects of Sound on Pupil Size and Variability



VERIFIED FINDINGS

- Initial fixation was slower for angry than happy faces, raw means 739.94 vs 702.28 ms, t(35) = 3.64, p = .003, indicating emotion-dependent differences in early visual orienting.
- Notification trials showed greater pupil variability than silence, 83.67 vs 79.80, t(35) = 2.99, p = .015. This is best interpreted as condition-dependent pupil instability, not a direct readout of a single cognitive state.
- Broader fixation, saccade, blink, and behavioral endpoints contained several null or context-dependent effects, reinforcing the value of multimodal rather than single-endpoint interpretation.

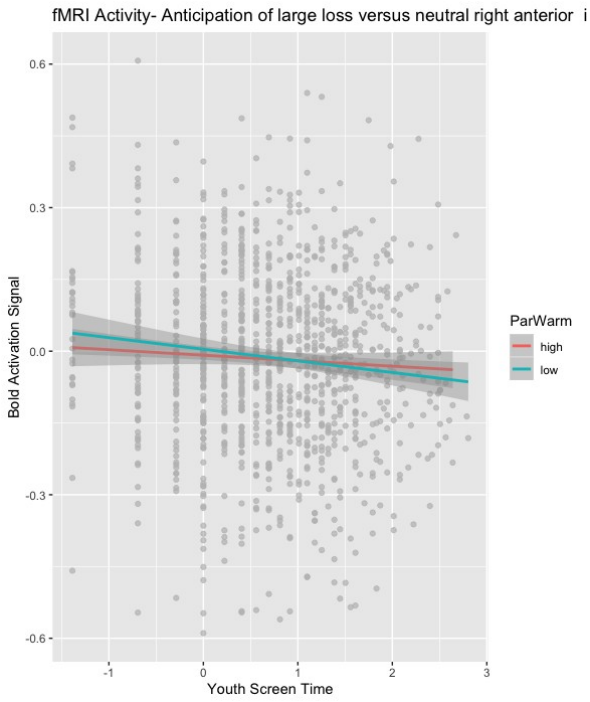
STUDY 4 | DEVELOPMENTAL NEUROIMAGING AND REWARD PROCESSING

Screen time, caregiver warmth, and fMRI reward circuitry

Graduate neuroimaging course project, 2024 | Secondary ABCD analysis, not peer reviewed

This project extended my research from millisecond-scale EEG and gaze measures to spatially localized fMRI outcomes. Using ABCD data, I tested whether screen time was associated with BOLD activity in reward and risk-processing regions during the Monetary Incentive Delay task, and whether perceived caregiver warmth moderated those relationships.

1,119	MID	NAcc + AIns
youth in analytic sample	reward anticipation task	predefined ROIs



Analysis design

- Youth screen time aggregated across multiple digital-media categories and log transformed.
- Caregiver warmth measured with the Acceptance Scale and transformed for skew.
- MRI motion >3 mm/s excluded; extreme BOLD beta values >3 SD excluded.
- Multiple regression in R/RMarkdown: ROI activity ~ screen time x caregiver warmth.

RESULT AND INTERPRETATION

Screen time was negatively associated with right anterior-insula activation during anticipation of large loss versus neutral, beta = -0.0322, p = .012. The screen-time by caregiver-warmth interaction was also significant, beta = 0.0000150, p = .047. The model explained little variance, adjusted R2 = .0073, so the result should be treated as a small observational association. A parallel right nucleus-accumbens interaction did not reach significance, p = .079.

What this added to my toolkit

The project added large-sample neuroimaging analysis, ROI-based BOLD interpretation, developmental covariate thinking, moderation modeling, and explicit attention to the difference between statistical significance and explanatory magnitude.

BEHAVIORAL, AFFECTIVE, AND CREATIVITY RESEARCH

Individual differences beyond the instrument

Peer-reviewed behavioral research and scholarly synthesis

My experimental neuroscience work sits within a broader program on attention, creativity, affect, mindfulness, and technology use. These projects strengthened psychometric measurement, individual-differences modeling, scientific writing, and the ability to connect laboratory measures with real-world behavior.

FIRST-AUTHOR STUDY: ICREATE

372	AUT + SemDis	CAQ
final participants	divergent-thinking originality	real-world creative achievement

Participants completed the Alternate Uses Task, semantic-distance scoring of divergent-thinking originality, the Creative Achievement Questionnaire, active/passive social-media measures, and a smartphone-addiction scale. Greater divergent-thinking originality was associated with less active social-media use and lower smartphone-addiction scores, while real-world creative achievement was associated with more active social-media use. Passive use was not associated with either form of creativity. The design was correlational, so causal claims were explicitly limited.

Publication

Upshaw, J. D., Davis, W. M., & Zabelina, D. L. (2022). iCreate: Social media use, divergent thinking, and real-life creative achievement. Translational Issues in Psychological Science, 8(1), 125. doi:10.1037/tps0000306

COLLABORATIVE AFFECTIVE SCIENCE

Imagination and anxiety during COVID-19

Two samples, N = 101 college students and N = 61 international community participants. More frequent imagination was associated with greater anxiety; loneliness amplified anxiety change among frequent imaginers. A valence interaction appeared in Study 1 but did not replicate in Study 2.

Attention, affect, and creativity

Co-authored Cambridge Handbook chapter synthesizing how focused, diffuse, and flexible attention interact with affect, mindfulness, mind wandering, and creative cognition. This work strengthened theory integration across behavioral and neuroscience literatures.

RESEARCH PATTERN

- Measure latent psychological constructs with multiple operationalizations rather than relying on one scale or endpoint.
- Use individual differences to test why the same perturbation may produce different behavior across people.
- Treat replication, null effects, and construct validity as part of the result rather than as problems to hide.

STUDY 5 | TRANSLATIONAL CLINICAL DATA SCIENCE AND AI

NSF SBIR Phase I: oncology prior-authorization workflow research

Principal Investigator | Retrospective, non-interventional, single-site technical feasibility study | 2025-2026

The Hidalga program translated the same measurement discipline into real-world clinical data. Instead of a tightly controlled laboratory task, the challenge was to reconstruct a partially observed healthcare workflow from heterogeneous EHR exports and determine which questions could be answered reliably before building prospective interventions.

24,167	200,513	148,703
age-eligible patients	analytic PA case records	free-text clinical notes

FIVE PRESPECIFIED RESEARCH QUESTIONS

H1 | Evidence extraction Can structured and free-text oncology records support reliable extraction of PA-relevant clinical evidence?	**H2 | Denial-risk triage** Can pre-decision information rank rare denial cases for additional staff review?	**H3 | Workflow timing** Where does turnaround-time burden occur, how does it vary by payer, and what next-action rules are plausible?
H4 | Trial-associated burden Do trial-associated PA cases show different rework, approval, appeal, or resubmission patterns?	**H5 | Operational value** What workflow and economic value could be plausible under explicitly modeled assumptions?	**Governance layer** What claims are retrospective, simulated, modeled, or ready for prospective testing?

ANALYTIC ARCHITECTURE

- Ten-stage pipeline covering cohort construction, data cleaning, longitudinal case reconstruction, clinical-field extraction, model development, calibration, workflow timing, sensitivity analysis, and reporting.
- Leakage controls and patient-level train/test separation for predictive modeling; PR-AUC used alongside ROC-AUC because denial prevalence was only about 2.4%.
- Data-quality review distinguished source-data, label, timing, and workflow-instrumentation problems from algorithmic underperformance.
- De-identified secondary data, Category 4 exempt human-subjects determination, with explicit provenance and claims-governance boundaries.

Research leadership

As NSF SBIR Principal Investigator, I was responsible for scientific validity, technical execution, model evaluation, clinical testing strategy, risk management, stakeholder collaboration, and transition planning. The project expanded my role from individual analyst to research-program leader.

STUDY 5 | RESULTS, LIMITATIONS, AND NEXT EXPERIMENT

From retrospective signal to prospectively testable system

The strongest outcome was not a single metric, but a defensible evidence chain for what to test next

0.967 F1	0.877 ROC-AUC	0.289 PR-AUC
combined clinical-field extraction	denial-risk discrimination	vs 0.024 denial baseline

WHAT THE RETROSPECTIVE DATA SUPPORTED

Evidence readiness
Only about 0.2% of cases met the defined complete PA-readiness criteria; the majority were partial or incomplete, identifying documentation readiness as a measurable intervention target.

Rare-outcome prediction
On 96,261 resolved cases, the selected LightGBM model exceeded the prespecified ROC-AUC target. Calibration and PR-AUC were used to define the model as review triage, not autonomous decision-making.

Payer timing
Strict-policy payer groups showed substantially longer observed turnaround time than lenient-policy groups, approximately 65 vs 35 median days in the reported comparison.

Workflow simulation
A calibrated simulation over 20,000 future PA cases estimated about 15% shorter median TAT and +4.5 percentage points approval rate. These are simulated, not observed clinical effects.

THE LIMITATION CHANGED THE NEXT STUDY

Historical records captured outcomes more reliably than the staff actions that produced them. That meant retrospective optimization could not establish a causal reduction in payer turnaround time. Rather than overstate the model, the project converted the missing information into a prospective telemetry specification: actual staff actions, eligibility, recommendation exposure, overrides and reasons, timestamps, payer responses, workload, and final outcomes.

Prospective pilot logic
The proposed Phase II sequence progresses from technical validation to silent evaluation and then staff-assisted workflow testing. The primary operational endpoint is a >=20% reduction in median chart-search and documentation-review time versus a specified comparator, with secondary measures of time to submit, rework, staff use, model accuracy, and payer outcomes.

TRANSFERABLE RESEARCH PRACTICE

1 | Define the phenomenon
Translate an ambiguous problem into measurable variables, cohorts, perturbations, and endpoints.

2 | Protect measurement validity
Treat signal quality, source authority, missingness, timing, and preprocessing as part of the experiment.

3 | Match inference to evidence
Separate association, prediction, simulation, intervention effect, and causal inference.

4 | Design the next experiment
Use nulls and limitations to specify the instrumentation and data that the next study must collect.

SELECTED WORKS REPRESENTED IN THIS PORTFOLIO

Upshaw, J. D., Stevens, C. E., Jr., Ganis, G., & Zabelina, D. L. (2022). The hidden cost of a smartphone: The effects of smartphone notifications on cognitive control from a behavioral and electrophysiological perspective. PLOS ONE, 17(11), e0277220.

Upshaw, J. D., Shields, G. S., Judah, M., & Zabelina, D. L. (2023). Electrophysiological effects of smartphone notifications on cognitive control following a brief mindfulness induction. Biological Psychology, 185, 108725.

Upshaw, J. D. (2025). Influence of Smartphone Notifications on Behavior and Eye Movement Parameters of Emotional Attention. Doctoral dissertation, University of Arkansas.

Upshaw, J. D., Davis, W. M., & Zabelina, D. L. (2022). iCreate: Social media use, divergent thinking, and real-life creative achievement. Translational Issues in Psychological Science, 8(1), 125.

Zabelina, D. L., Clay, J. Z., & Upshaw, J. D. (2021). The association between imagination and anxiety in the times of the COVID-19 pandemic. Creativity Research Journal, 33(3), 264-274.

Kane, S. J., Awa, K. N., Upshaw, J. D., Hubert, K., Stevens, C. E., Jr., & Zabelina, D. L. (2023). Attention, affect, and creativity, from mindfulness to mind-wandering. In The Cambridge Handbook of Creativity and Emotions.

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